

COMPARATIVE SQL CASE STUDY

Two Seasons, One Strategic Truth

An executive relational database deep dive comparing player optimization, roster profiles, and matching architectures across the 2018 and 2019 tournament runs.

Data Strategist: **Aditi Paitandy**

Platform Integration: Google BigQuery Engine

| The Search for Tactical Efficiency

COGNITIVE BIAS

The Overvaluation Trap

Sports managers consistently make multi-million dollar errors by relying on aggregate gross totals (like seasonal runs or wickets) while ignoring fundamental unit efficiencies.

⚠ Consequence: Inefficient Capital Deployment

OBJECTIVE

The Core Analytical Goal

Our objective is to employ relational models that isolate true strategic performance, neutralize sample outliers, and trace team composition rules across cohorts.

✅ Solution: Production-Grade SQL Parsing

| Architecting the Database Pipeline



Schema Design

Google BigQuery tables isolated by year and role. Mapped relational schema keys across multi-season cohorts with consistent datatypes.



Workflow Integrity

Calculated dynamically computed database metrics directly in-query (e.g. converting overs and runs conceded into verified Economy Rates).



Rigorous Filtering

Standardized join relationships and applied logical mathematical thresholds to omit low-sample anomalies from top-level metrics.

Roster Design: Battling Squad Size

To understand team configurations, we grouped the active batsman rosters across franchises for the 2019 season.

```
-- Calculate batsman volume per franchise
SELECT Team, COUNT(Player) AS TotalCount
FROM `men_IPL.2019_Batsmen`
GROUP BY Team;
```

Discovery: Championship organizations consistently lock down structured, tight squads of exactly 5 to 7 batting specialists.

Core Roster Sizes (2019)



| Debunking the Chase-First Bias

54%

2018 CHASE WIN RATIO

48%

2019 CHASE WIN RATIO

Chasing Was Not an Instant Cheat Code

Many franchises operate under the cognitive bias that chasing matches delivers a significant tactical advantage. Our SQL comparative study proved that the "chasing bias" evaporated in 2019.

As pitch conditions degraded and defensive bowling anchors adjusted, teams defending targets gained the edge—making a strict "chase-first" toss selection a liability.

The Strategic Defensive Anchors

True defensive giants combine strike rate power with tight economy rates under pressure (Economy Rate < 7.00).

```
SELECT Player, Wkts, E_R
FROM `men_IPL.2018_Bowlers`
WHERE E_R < 7
ORDER BY Wkts DESC;
```

PLAYER	WICKETS	ECONOMY (E_R)
Rashid Khan	21	6.74
Bumrah, J J	17	6.89
Mujeeb Ur Rahman	14	6.99
Ngidi, L	11	6.00

Unlocking Multi-Year Consistency

MULTI-SEASON INNER JOIN

The Run-Scoring Champion

Single-season success often skews player valuation. To find true long-term stability, we executed an inner join across both seasons on the player key.

```
SELECT a.Player, (a.Runs + b.Runs) AS CombinedRuns
FROM `2018_Batsmen` a
INNER JOIN `2019_Batsmen` b ON a.Player = b.Player
ORDER BY CombinedRuns DESC LIMIT 1;
```

CONSISTENCY KING

K L Rahul

TOTAL SUM: 1,252 RUNS

659 Runs

593 Runs

2018 Season

2019 Season

The Outlier Hazard: Volume Bias

NAIVE QUERY BIAS

Why Unfiltered Metrics Deceive

A simple query seeking "best bowlers with lowest runs conceded" returned Yusuf Pathan (14 runs conceded) and D'Arcy Short (19 runs conceded).

But auditing further reveals these players only bowled 2.0 and 3.0 overs respectively—skewing standard performance parameters completely.

✗ Omitted: Workload Minimum Floor

CORRECTED LOGICAL QUERY

Rigorous Threshold Solution

By enforcing logical workflow constraints (such as minimum workload filters like `Overs >= 10.0`), we isolate true high-value assets and filter out small-sample noise.

In our final corrected query, true defensive anchors like Ish Sodhi (Economy: 5.87) rise to the top.

✓ Applied: `Overs ≥ 10.0 Floor`

| The Production-Grade SQL Engine



Relational Joins

Experienced in structuring database maps across multi-year schemas, handling potential data drifts cleanly.



Advanced Aggregations

Capable of computing grouped aggregates, subqueries, and window operations to transform raw tables into business-ready answers.



Strategic Translation

Skilled in translating complex code logic into clear metrics that drive roster, corporate, and operational growth decisions.

AVAILABLE FOR DATA ANALYST ROLES

Let's build something exceptional together

Hi, I'm **Aditi Paitandy**. I specialize in querying databases, building pipelines, and transforming raw structured data into high-stakes strategic answers.



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